

Namba & Dotonbori canals, Osaka

Then-and-now aerial photographs, how the ground was made, and the disasters people chose to remember — compiled automatically from public Japanese government data.



難波・道頓堀（大阪市中央区・浪速区）

Study area	2.4 km × 1.5 km around 34.6685° N, 135.4960° E
Oldest aerial photograph	c. 1936–1942
Aerial series in this brief	c. 1928, c. 1936–1942, 1945–1950, 1961–1969, 1974–1978, 1979–1983, 1984–1986, latest
Main natural landform	Sandbars and dunes (61%)
Former river channels	none mapped
Filled, reclaimed or drained land	none mapped
Former sea, river or pond	none mapped
Meiji-era paddies, marsh and flats	none mapped
Disaster memorials within 3 km	4 (storm surge 2, other 2, earthquake 2, tsunami 2)

What this brief is. A reading aid for comparing the same ground across decades. It shows what the photographs and GSI's land surveys record; it does not describe current conditions, access or opening hours.

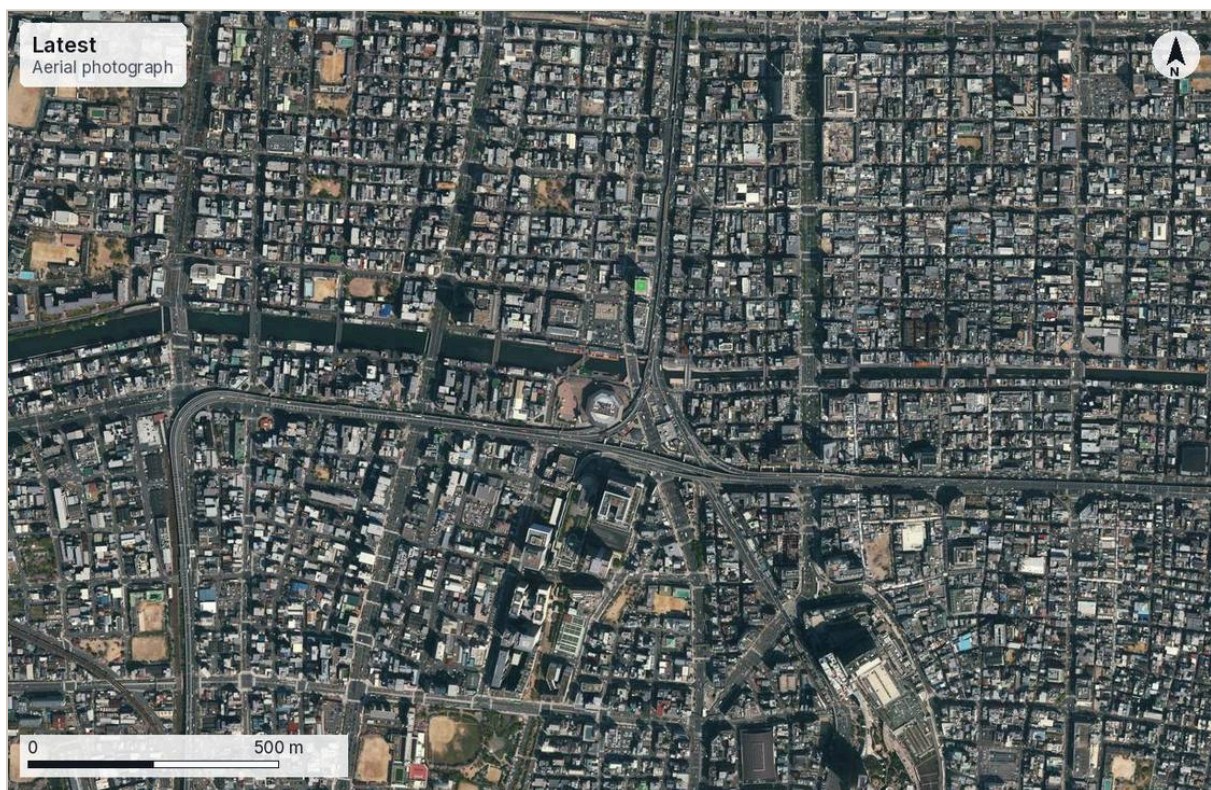
What it is not. Not a hazard assessment or a property survey. Landform classes describe typical tendencies of the ground; for present-day risk, read the official hazard map of the municipality.

Edition 2026-09-27. Aerial photographs, landform and Meiji-era lowland data: Geospatial Information Authority of Japan (GSI), processed by Japan Time Atlas. Memorial data: GSI Natural Disaster Memorials, summarised and translated by Japan Time Atlas.

The same ground, c. 1936–1942 and today



c. 1936–1942 aerial photograph — GSI Tiles *ort_riku10*, cropped and annotated by Japan Time Atlas. North is up; the frame is identical in both images.



Latest seamless mosaic — GSI Tiles *seamlessphoto*, cropped and annotated by Japan Time Atlas. Capture dates within a mosaic can differ from place to place.

How to compare. Anchor on features that rarely move — river banks, rail lines, shrine and temple grounds, main roads — then look for what changed around them. Differences in tone come from film, camera and season, not from the land itself.

The decades in between

Every further GSI series that covers the same frame, oldest first. Each series is a compilation: photographs within one series were taken on different dates, which is why seams and changes of tone can appear inside a frame.



c. 1928 · GSI Tiles *ort_1928*, processed



1945–1950 (US military photography) · GSI Tiles *ort_USA10*, processed



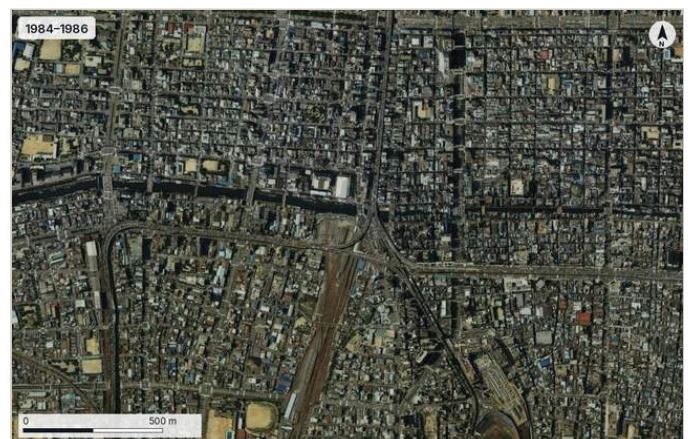
1961–1969 · GSI Tiles *ort_old10*, processed



1974–1978 · GSI Tiles *gazo1*, processed

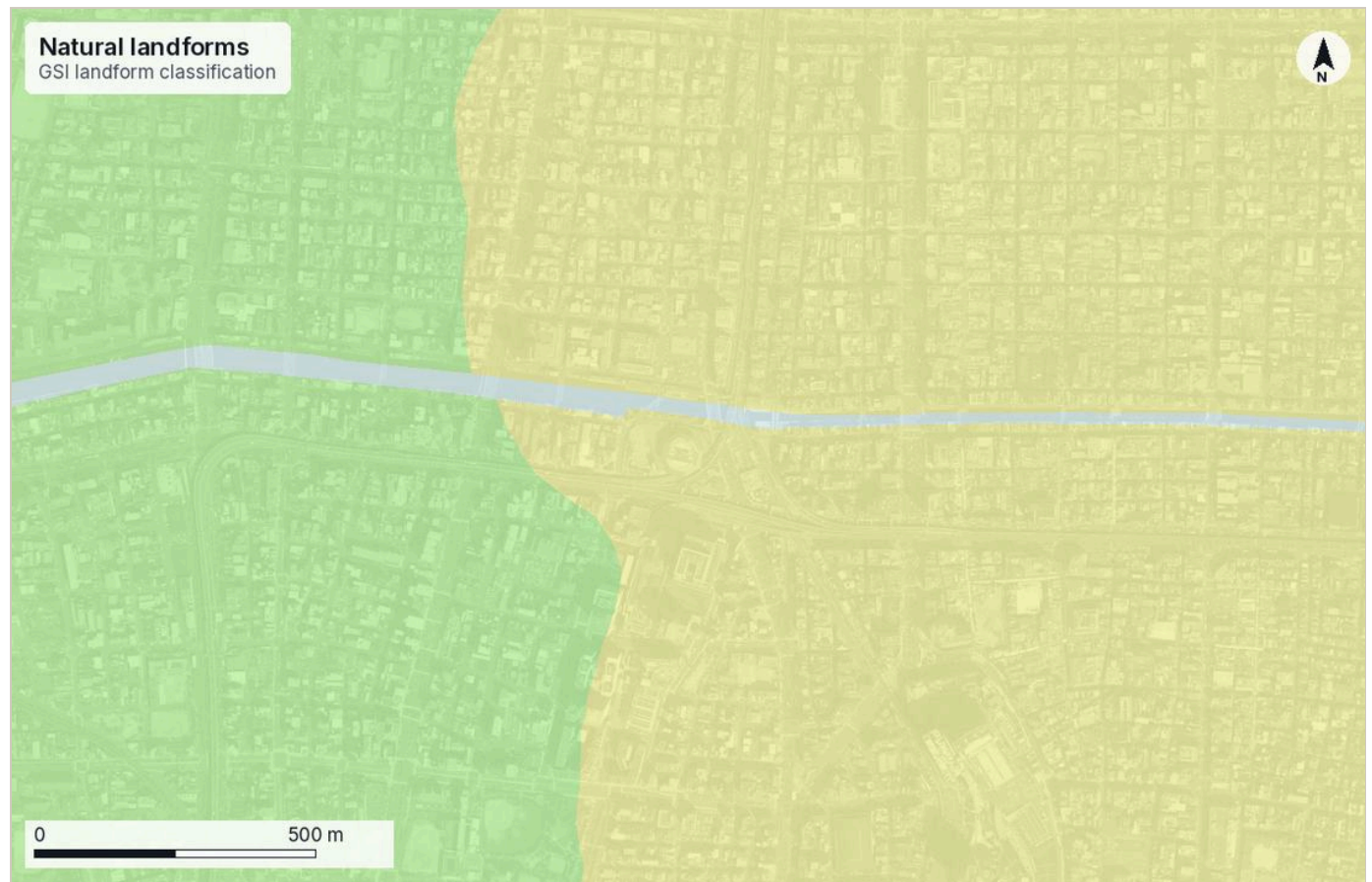


1979–1983 · GSI Tiles *gazo2*, processed



1984–1986 · GSI Tiles *gazo3*, processed

Rivers, shorelines and terraces



Natural landforms as classified by GSI, over a recent aerial photograph in gray. Dark blue outlines mark former river channels.

- Sandbars and dunes 61%
- Floodplain and coastal plain 37%
- Water 2.0%

What the main landforms mean

Sandbars and dunes (61%). Sandy or gravelly ground higher than its surroundings along present or former coasts, lakeshores and riverbanks, built by waves or wind. Usually escapes ordinary floods. Liquefaction tendency is somewhat strong on sandbars, somewhat weak on dunes and very strong at dune margins.

Floodplain and coastal plain (37%). Low, flat ground of little relief, built from sand and mud left by floods, or former seabed that emerged. River flooding. Ground is softer nearer the coast and shakes somewhat more; liquefaction tendency is strong on coastal plains and somewhat strong on floodplains. Storm surge near the coast.

Descriptions translated and condensed from GSI's landform-classification legend. They describe tendencies of each landform type, not the condition of any individual plot.

Fill, reclamation and the fields beneath the town



Hatching: man-made ground as classified by GSI. Teal outlines: former sea, river or pond. Dark blue outlines: former river channels.



GSI “Meiji-era lowland” data, traced from topographic maps of the 1880s–1900s; GSI warns that positions can be off by a considerable distance, especially in the Kanto and Kinki regions. Only low, wet or open ground was traced: towns, dry fields and forest of the period are left blank.

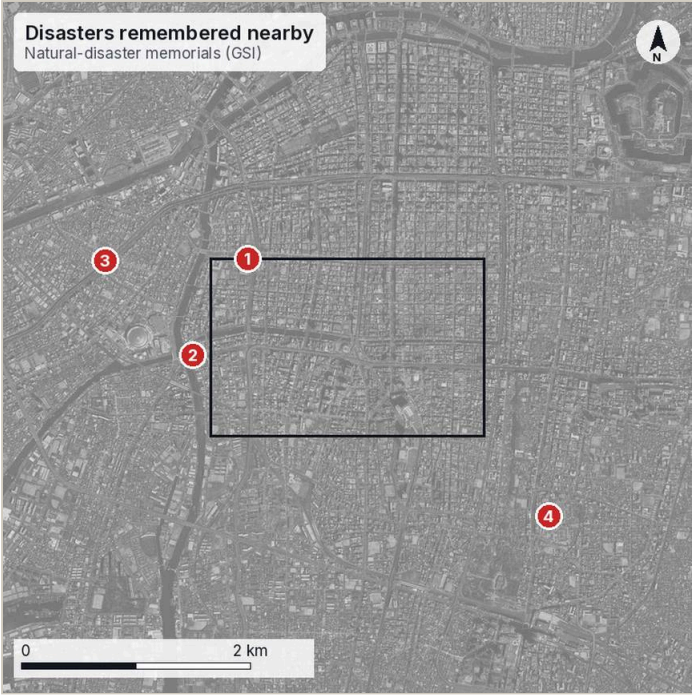
Man-made ground

No man-made ground mapped in this frame.

Meiji-era lowland

■ River, pond or sea 5.8%

Memorials to natural disasters nearby



Numbered memorials; the black rectangle is the study area of this brief. Frame 6 km across.

Memorial	Disaster	Erected	Distance
1 Flood defense monument recalling the storm surges of 1934, 1950 and 1961 水防碑	Muroto Typhoon (21 Sep 1934), Typhoon Jane (3 Sep 1950) and Second Muroto Typhoon (16 Sep 1961) · storm surge / other	1978	1.2 km
2 Stone recording the 1854 Ansei earthquake and tsunami at the two river mouths 大地震両川口津浪記	Ansei Tokai earthquake (23 Dec 1854) and Ansei Nankai earthquake (24 Dec 1854) · earthquake / tsunami	1855	1.4 km
3 Stone record of the 1934 Muroto Typhoon storm and flood in Kujo 暴風水害記念誌	Muroto Typhoon (21 Sep 1934) · storm surge / other	1938	2.2 km
4 Monument to the 1854 Ansei earthquake and tsunami 安政地震津波碑	Ansei Tokai earthquake (23 Dec 1854) and Ansei Nankai earthquake (24 Dec 1854) · earthquake / tsunami	1855	2.3 km

What the memorials record

1. Flood defense monument recalling the storm surges of 1934, 1950 and 1961. Nishi Ward in Osaka has been flooded again and again by storm surges brought by typhoons. Many lives and much property were lost: in Osaka City, 990 people died in the Muroto Typhoon of 1934, 211 in Typhoon Jane of 1950 and 6 in the Second Muroto Typhoon of 1961.
2. Stone recording the 1854 Ansei earthquake and tsunami at the two river mouths. After the Ansei Nankai earthquake of 24 December 1854, a tsunami struck people who had taken refuge in boats moored on the Aji, Kizu and other rivers, with heavy losses. The stone was erected as a warning to later generations, because the lesson of a similar disaster in the 1707 Hoei earthquake had not been heeded.
3. Stone record of the 1934 Muroto Typhoon storm and flood in Kujo. The Muroto Typhoon hit Osaka at about 8 am on 21 September 1934 with winds of 60 m per second and a storm surge of nearly 3 m. The city's dead and missing reached 990; 28 wooden school buildings collapsed and 269 pupils died. In Kujo, roof tiles flew like leaves and every house was flooded above floor level.
4. Monument to the 1854 Ansei earthquake and tsunami. People who took to small boats to escape the earthquakes of 23 and 24 December 1854 were swept upstream, boats and all, by a tsunami that surged into the mouth of the Kizu River, and a great many died. The inscription warns that a rumbling sea or disturbed tides are signs of a tsunami, and that people should evacuate.

Source: GSI Natural Disaster Memorials (自然災害伝承碑). English names and notes are Japan Time Atlas translations and summaries of GSI's Japanese descriptions; the memorials' own inscriptions may say more. "Erected" is the year GSI records for the monument.

Where every element comes from

- **Aerial photographs** (pp. 2–3) — Geospatial Information Authority of Japan (GSI), GSI Tiles: series *ort_1928* (c. 1928), *ort_riku10* (c. 1936–1942), *ort_USA10* (1945–1950, US military photography), *ort_old10* (1961–1969), *gazo1* (1974–1978), *gazo2* (1979–1983), *gazo3* (1984–1986) and *seamlessphoto*. Cropped, resized and annotated by Japan Time Atlas. List of GSI Tiles: <https://maps.gsi.go.jp/development/ichiran.html>
- **Landform classification** (pp. 4–5) — GSI landform-classification vector tiles (natural and artificial landforms), compiled by GSI from its Land Condition Maps, Flood Control Terrain Classification Maps and vulnerable-landform surveys. Rendered, measured and translated into English by Japan Time Atlas. https://github.com/gsi-cyberjapan/experimental_landformclassification
- **Meiji-era lowland** (p. 5) — GSI “明治期の低湿地” tiles and legend (https://maps.gsi.go.jp/legend/lw_legend.pdf), measured and translated by Japan Time Atlas.
- **Natural disaster memorials** (p. 6) — GSI Natural Disaster Memorials (自然災害伝承碑), registered by municipalities, <https://www.gsi.go.jp/bousaichiri/denshouhi.html>; processed by Japan Time Atlas (selection by distance, English translation and summary). Memorial photographs are not reproduced.
- GSI content is used under the GSI Content Terms of Use, which follow the Public Data License v1.0 (PDL1.0, compatible with CC BY 4.0). Data retrieved 2026-09-27.
- **Outline of Japan** — Data of Japan / JapanPrefGeoJson (public domain), simplified.

Method

The study area is a fixed rectangle of 2.4 × 1.5 km. Every GSI photo series that covers at least 85% of it is included; the newest annual orthophoto covering at least 90% of it is used as “now”, with any gap filled from GSI's latest seamless mosaic. Landform shares are areas of the rectangle measured on GSI's polygons at zoom level 14, each class merged before measuring. Meiji-era shares count map pixels by legend color. Memorials are selected by straight-line distance from the center of the study area.

Limits

Historical series are mosaics of photographs taken on different dates; GSI publishes the flight date of each photograph. Landform data record conditions at the time of survey. Nothing in this brief shows present-day access, safety or opening hours.

Area Dossier. The same district with a 4–8 page English history, written from Japanese library sources and cited page by page.

Deep Research. A single hamlet or address, researched in libraries and archives.

Compiled by Japan Time Atlas; not produced, reviewed or endorsed by GSI, MLIT or any Japanese government agency. Not an official hazard map and not for statutory real-estate disclosure. Generated 2026-09-27 by an automated pipeline (research/lab-07/scripts/basic.py).